

Roof Lifecycle Cost Analysis

Culbertson (N. Virginia) market
October 2, 2026

Planned maintenance is projected to save the owner of a typical Culbertson (N. Virginia) mixed-use building **\$266K** over the next 20 years.

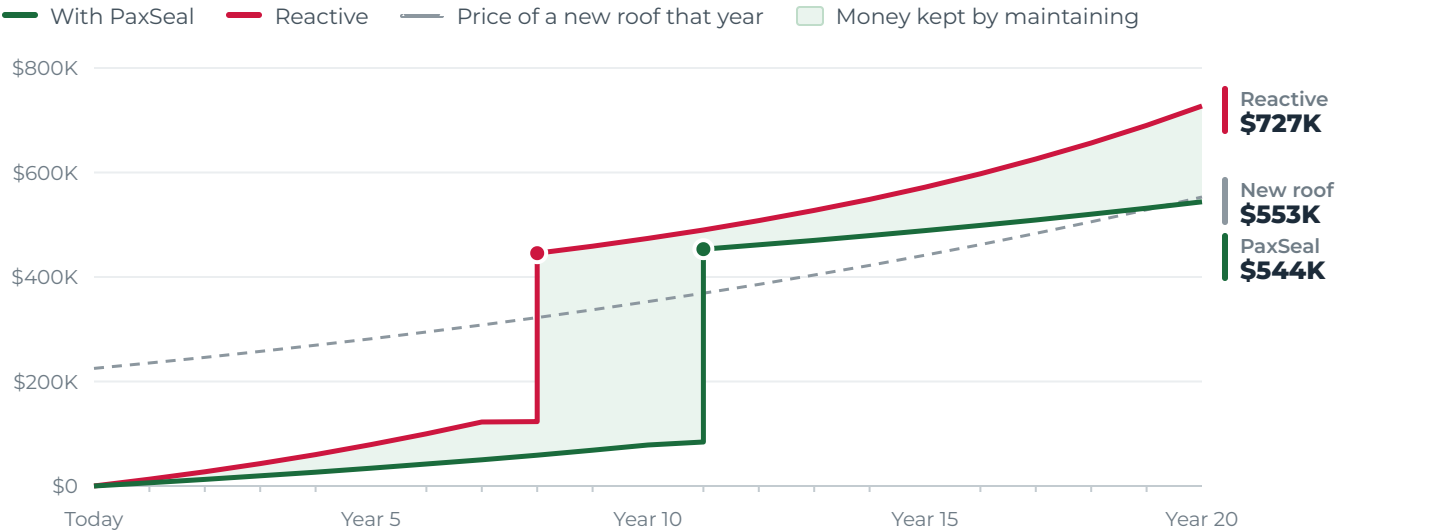
That is 31.9% less than fixing problems only after they leak. Under conservative to upside assumptions, the savings range from \$125K to \$356K. In today's dollars the estimate is worth \$140K.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE	NEW ROOF TODAY
Culbertson (N. Virginia)	Mixed Use	25,000 sq ft	TPO	8 yrs Good	\$225K

PAXSEAL NET COST \$566K \$544K spent plus roof wear, net of roof life left	REACTIVE NET COST \$832K \$727K spent plus roof wear, net of roof life left	VALUE PROTECTED \$149K \$10K/yr more NOI at a 6.8% cap rate	NEW ROOF PRICE 2× in 15 yrs \$225K today, \$553K by year 20
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Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal program fees	\$109K	—	-\$109K
Repairs paid by owner	\$55K	\$388K	+\$333K
Expected storm damage	\$11K	\$17K	+\$6K
Roof replacements	\$369K	\$322K	-\$47K
Current roof value used	\$135K	\$135K	—
Credit for roof life left	-\$113K	-\$30K	+\$83K
Net 20-year cost	\$566K	\$832K	+\$266K

What this means

- 1 **A later replacement.** The next new roof moves from year 8 to year 11, when it costs less in today's dollars.
- 2 **Smaller repairs.** Fixes made on scheduled visits avoid emergency work at about 3.6× the cost in a mixed-use building, saving \$333K.
- 3 **Both income streams protected.** A leak can close the ground-floor retail and damage the offices or apartments above.
- 4 **The program pays for itself.** Each \$1 of fees avoids about \$3.44 of other roof costs.

Local risk profile

Prince William County, VA (51153)

Thunderstorm wind / derecho, snow and ice

Strong wind and winter weather both rated Relatively High (FEMA NRI); tropical remnants ~1 per 12 years. Expected storm damage to this roof over 20 years is **\$11K** if maintained and **\$17K** if neglected. A documented maintenance history also helps when an insurer questions a claim.

Hail	Moderate	
Strong wind	High	
Tornado	Moderate	
Hurricane	Moderate	
Winter weather	High	
Ice storm	Low	

FEMA National Risk Index rating

Key assumptions

The basis for each figure in this analysis.

ASSUMPTION	VALUE	BASIS
Roof life	20 / 15 yrs	Maintained / reactive. Fannie Mae service-life tables; +5 yrs per NRC Canada and warranty terms
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing contractors, 2007–2025
Replacement cost	\$9.00/sq ft	Culbertson (N. Virginia) contractor price ranges
Reactive repair cost	3× planned × 1.2	Industry estimates; no primary study found. ×1.2 for leak damage in a mixed-use building
Storm damage	0.15%/yr	Of roof cost, before insurance. FEMA National Risk Index expected building loss, roof-related hazards
PaxSeal fee	\$4,050/yr	PAX program estimate for this roof size. Your proposal will state the actual fee
Cap rate	6.8%	Blend of 30% retail, 35% office and 35% multi-family rates until local data is added

Estimates, not a quote. Projections from Culbertson (N. Virginia) market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20. Storm damage is before insurance. Property value is the yearly gain in net operating income divided by the cap rate, for income property where the owner pays for the roof.